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SECP3843-02 SPECIAL TOPIC IN DATA ENGINEERING

Tutorial Article

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1.0 Main problem

PT. MPM, a regional packaging manufacturer in Indonesia, lacks a centralized and appropriate application or system to accurately measure, track, and monitor the performance of its critical core processes (Jayadinata et. al, 2024). Currently, operational data is scattered across separate reports (such as production and maintenance logs). Because of this data fragmentation, top management is forced to make business decisions based on manual, lagging info. This lack of real-time visibility into Key Performance Indicators (KPIs) reduces operational efficiency and lowers the company's competitiveness against its market rivals.

2.0 Main solution

The solution developed by Jayadinata et. al, 2024, was a web-based Business Intelligence (BI) performance monitoring dashboard powered by Google Data Studio, built using the structured, linear Waterfall methodology. The solution breaks down into two interactive dashboards:

1. **Sales Performance Dashboard:** Compares actual revenue versus forecasts, tracking customer and product performance using heatmaps and combo charts.
2. **Production Performance Dashboard:** Tracks machine-level productivity and gauges Overall Equipment Effectiveness (OEE) factors like availability, performance and quality ratios.

3.0 Use Case Diagram

The use case diagram for the PT. MPM BI Dashboard System involves six actors: Top Management, Data Analyst, Tech & Ind. Design, Inventory & Warehouse, Sales Team and Manufacturing Team. Top Management has the broadest access which interacts with all major use cases including Login/Access Dashboard, View Sales Dashboard, View Production Dashboard and Share & Export Report. The Data Analyst is responsible for the data pipeline by performing Upload & Update Data which mandatorily includes the Run ETL Process via and <<include>> relationship. Moreover, Tech & Ind. Design handles the Register/Update Product Info use case, ensuring product data remains accurate within the system. While Inventory & Warehouse have both access to the View Sales Dashboard and Production Dashboard, the Sales Team and the Manufacturing Team can only access View Sales Dashboard and View Production

Dashboard use cases respectively. Additionally, both Sales and Manufacturing team have access to Share & Export use cases. All actors share the Login/Access Dashboard use case as the entry point to the system (Jayadinata et. al, 2024).

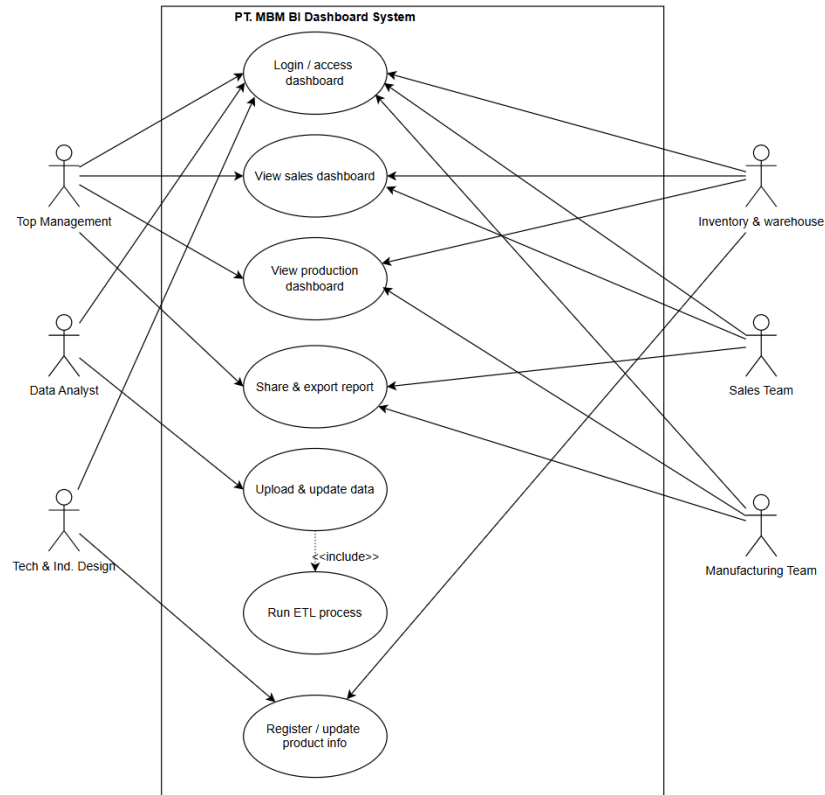


Figure 3.1 Use case diagram

4.0 User Story

User stories are utilized to capture the functional requirements of the PT. MPM BI Dashboard System from the perspective of its various stakeholders . By defining what each user role needs and the underlying business value, these stories guide the development team in building relevant dashboard features such as tracking KPIs, drill-down filtering and automated data pipelines . The specific requirements for each user group are detailed below:

Table 4.1 captures the high-level strategic requirements for Top Management who need comprehensive visibility across the organization to make data-driven, long-term business decisions.

Table 4.1 : User story of Top Management

US No	User Story
US-01	As a Top Management, I want to login and access the dashboard, so that I can securely view the company's overall performance data
US-02	As a Top Management, I want to view the sales dashboard, so that I can monitor revenue and evaluate whether sales KPIs are being met
US-03	As a Top Management, I want to view the production dashboard, so that I can oversee overall production performance across all divisions
US-04	As a Top Management, I want to share and export reports, so that I can distribute performance summaries to relevant stakeholders

Table 4.2 describes the technical data management obligations of the Data Analyst, focusing on maintaining accurate, clean, and up-to-date data structures within the data warehouse.

Table 4.2 : User story of Data Analyst

US No	User Story
US-05	As a Data Analyst, I want to login and access the dashboard, so that I can verify that data is being displayed correctly
US-06	As a Data Analyst, I want to upload and update data, so that the dashboard always reflects the latest sales and production figures
US-07	As a Data Analyst, I want to run the ETL process automatically when uploading data, so that raw data is cleaned, transformed and loaded properly into the data warehouse

Table 4.3 outlines the product management needs of the Tech & Industrial Design staff, who are responsible for registering and updating product information to ensure the system always reflects the latest product specifications.

Table 4.3 : User story of Tech & Industrial Design

US No	User Story
US-08	As a Tech & Industrial Design staff, I want to login and access the dashboard, so that I can view product and production related information
US-09	As a Tech & Industrial Design staff, I want to register and update product information, so that the latest product specifications are recorded accurately in the system

Table 4.4 reflects the operational needs of the Inventory & Warehouse staff, who require dashboard access to monitor stock levels, track production output, and align inventory with sales demand and product information.

Table 4.4 : User story of Inventory & Warehouse

US No	User Story
US-10	As an Inventory & Warehouse staff, I want to login and access the dashboard, so that I can monitor stock and product availability
US-11	As an Inventory & Warehouse staff, I want to view the sales dashboard, so that I can understand sales demand and align inventory levels accordingly
US-12	As an Inventory & Warehouse staff, I want to view the production dashboard, so that I can track production output and manage stock replenishment
US-13	As an Inventory & Warehouse staff, I want to register and update product information, so that the latest stock and product details are accurately recorded in the system

Table 4.5 captures the sales monitoring requirements of the Sales Team, who need dashboard access to track sales progress against targets and share performance updates with relevant stakeholders.

Table 4.5 : User story of Sales Team

US No	User Story
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US-14	As a Sales Team member, I want to login and access the dashboard, so that I can view sales performance relevant to my division
US-15	As a Sales Team member, I want to view the sales dashboard, so that I can track sales progress against targets for my assigned accounts
US-16	As a Sales Team member, I want to share and export reports, so that I can share sales performance updates with my team or management

Table 4.6 outlines the production monitoring needs of the Manufacturing Team, who require dashboard access to track actual output against planned targets, update production data, and distribute performance reports to relevant stakeholders.

Table 4.6 : User story of Manufacturing Team

US No	User Story
US-17	As a Manufacturing Team member, I want to login and access the dashboard, so that I can monitor production performance for my division
US-18	As a Manufacturing Team member, I want to view the production dashboard, so that I can track actual output against planned production targets
US-19	As a Manufacturing Team member, I want to share and export reports, so that I can distribute production performance updates to relevant stakeholders

5.0 Entity Relationship Diagram

Figure 5.1 illustrates the Entity Relationship Diagram (ERD) of the PT. MPM Business Intelligence Dashboard System, which consists of seven entities: CUSTOMER, SALES_ORDER, SALES_ORDER_LINE, PRODUCT, PRODUCTION_RECORD, MACHINE, and OPERATOR.

On the sales side, CUSTOMER is linked to SALES_ORDER through a "places" relationship, where one customer can place one or more orders. Each SALES_ORDER is then broken down into one or more SALES_ORDER_LINE records, with each line item referencing

exactly one PRODUCT. This captures the full sales transaction from customer order down to individual product level.

On the production side, PRODUCT is linked to PRODUCTION_RECORD through a "produced in" relationship, connecting the sales and production sides of the business. Each PRODUCTION_RECORD logs detailed production activity including planned versus actual output, run time, stop time, and OEE metrics such as availability, performance, and quality. MACHINE and OPERATOR are both linked to PRODUCTION_RECORD, recording which machine was used and which operator was assigned for each production run.

Overall, the ERD provides the data foundation for both the sales performance dashboard and the production performance dashboard of the PT. MPM BI system.

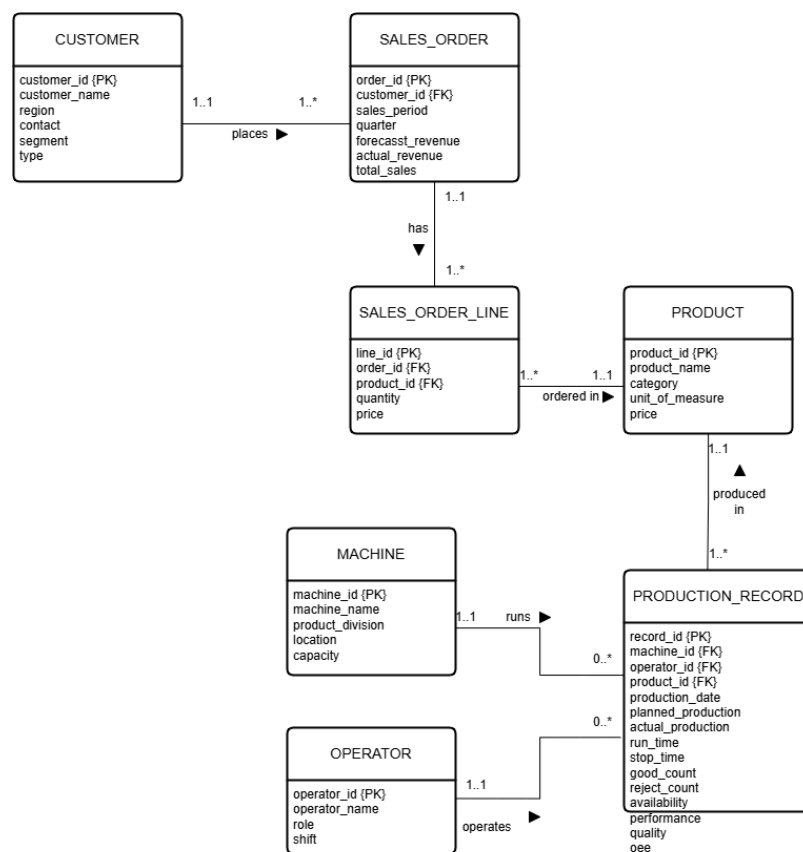


Figure 5.1 : Entity Relationship Diagram (ERD)

6.0 Data Architecture

The data and system architecture of the PT. MPM Business Intelligence (BI) Dashboard follows a four-layer pipeline model. This architecture aligns with the Waterfall methodology adopted by Jayadinata et al. (2024), ensuring a structured and traceable data flow from raw operational data to management insights.

As shown in Figure 6.1, Layer 1 consists of the source systems, where raw data is collected from four operational systems: the ERP System (production reports), the Sales System (orders and forecasts), the Inventory System (product stock), and the Maintenance System (machine records). In Layer 2, the data undergoes the Extract, Transform, and Load (ETL) process, where raw reports are converted into Google Sheets format, cleaned, and loaded into two data marts which are the Sales Data Mart and the Production Data Mart. Layer 3 represents the BI presentation layer built using Google Data Studio, producing two interactive dashboards including the Sales Performance Dashboard and the Production Performance Dashboard accessible via web browser. Finally, Layer 4 is the end-user access point, where PT. MPM's top management from Operations, Finance, Production, and Director level interact with the dashboards to support data-driven decision-making.

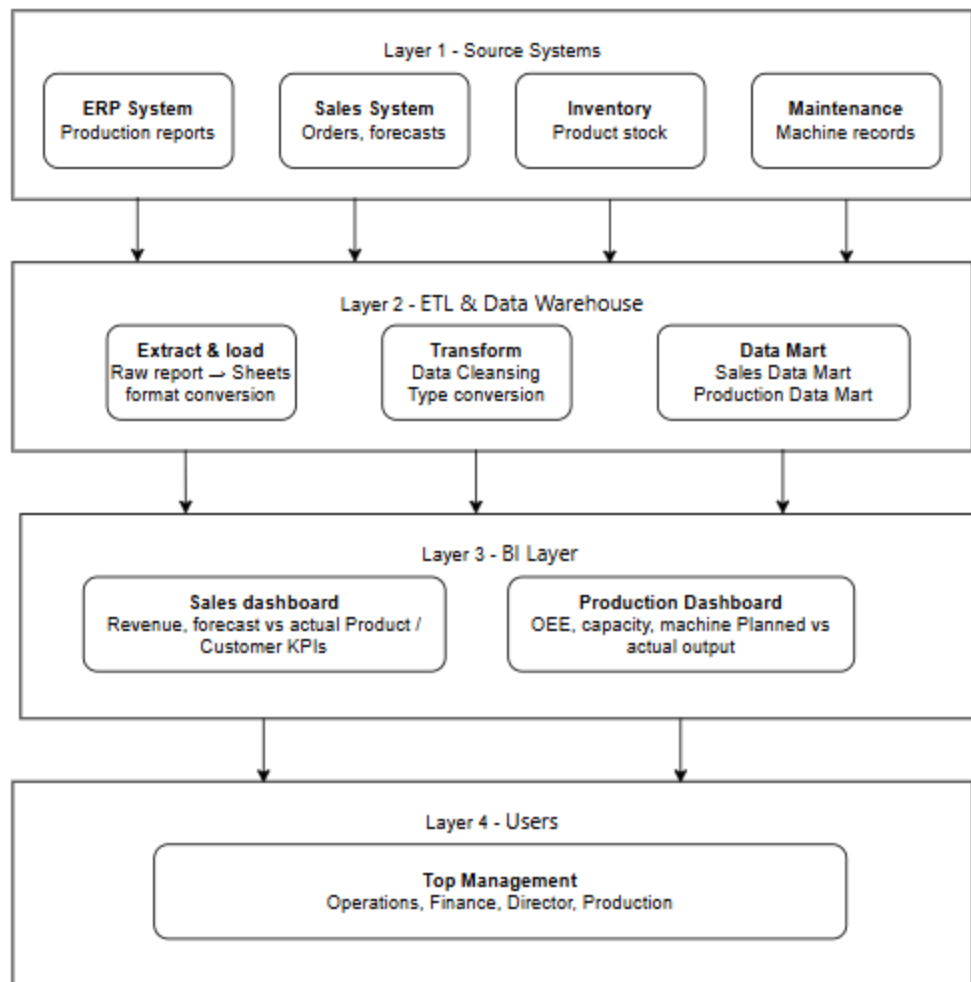


Figure 6.1 : Data Architecture

6.0 Multidimensional Star Schema Design

As illustrated in Figure 6.1, the multidimensional star schema for PT. MPM's BI Dashboard is designed as a galaxy schema consisting of two fact tables, FACT_SALES and FACT_PRODUCTION, supported by five dimension tables. Both fact tables share DIM_DATE and DIM_PRODUCT as conformed dimensions using the surrogate key pattern (integer _key columns), enabling cross-domain analysis between sales and production data (Sherman, 2014).

FACT_SALES captures total_sales, forecast_sales, revenue, and forecast_revenue, and is connected to DIM_CUSTOMER (customer_key, customer_id, customer_name, customer_type, region, created_at), DIM_DATE (date_key, full_date, month, quarter, year, week_number, shift, period_label), and DIM_PRODUCT (product_key, product_id, product_name, product_category, product_division, unit_price, uom). FACT_PRODUCTION captures planned_production, actual_production, good_count, reject_count, run_time, stop_time, and oee as a pre-computed DECIMAL(5,2) value, and is connected to the shared DIM_DATE and DIM_PRODUCT, as well as DIM_MACHINE (machine_key, machine_id, machine_name, machine_type, capacity, location, status) and DIM_OPERATOR (operator_key, operator_id, operator_name, department, role, shift). The OEE value is derivable at query time as $\text{Availability} (\text{run_time} / \text{planned_production_time}) \times \text{Performance} (\text{actual_production} / (\text{planned_production} \times \text{run_time})) \times \text{Quality} (\text{good_count} / (\text{good_count} + \text{reject_count}))$, as visualised through gauge charts in the Production Performance Dashboard (Jayadinata et al., 2024).

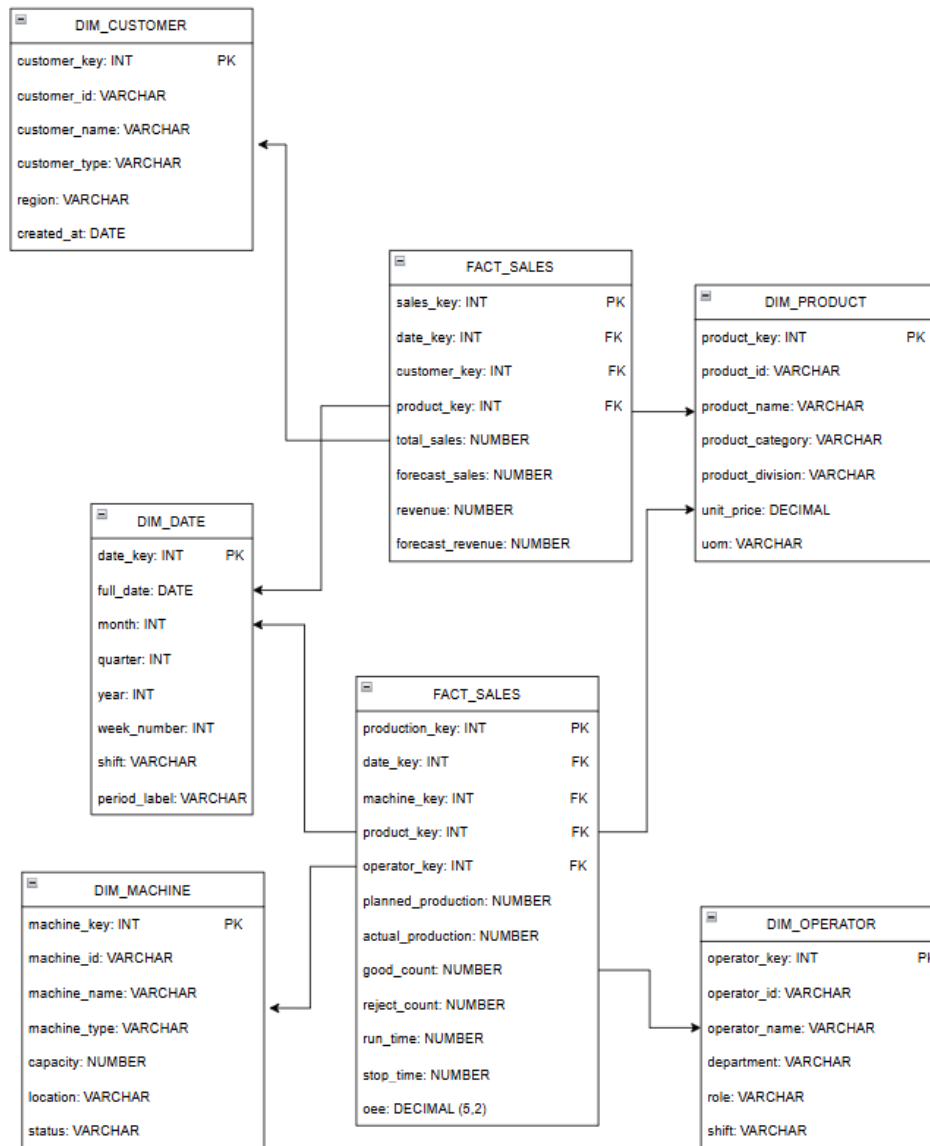


Figure 6.1 : Galaxy Schema

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- Jayadinata, D. H., Pratama, A. T., & Sofianti, T. D. (2024). Development of performance monitoring dashboard for product packaging manufacturer by using waterfall methodology. *Jurnal Ilmiah Teknik Industri*, 23(1), 151–161. <https://doi.org/10.23917/jiti.v23i1.2551>
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